Verification plan & Subroutines – Extra

1)

Verify the functionality of the following D-FF:

- **Inputs:** clk, rst (active high sync), d, en
- **Outputs:** q
- **Parameters:** USE_EN (if equals 1 then use the enable to update the q output when the reset is deasserted, else ignore the en input)

Since the design has a parameter, we need to verify the functionality of both modes when the USE_EN equals 1 and zero.

Requirements:

- Create a verification plan -> Use this verification plan as a template
- You should create 2 self-checking testbenches (dff_t1_tb.sv & dff_t2_tb.sv) to verify the functionality of both parameter configuration.
- Use exhaustive test patterns where all combinations of inputs are tested.
- Use a task to check the functionality of the design.
- Compute the expected output value to compare the DUT output or create a golden model (always block to generate the correct expected functionality) inside of the testbench to compare the DUT output to your reference output.
- Use a do file
    - Compile the design and 2 testbenches
    - Run simulation for the first testbench
    - Close the simulation (quit -sim) to save the coverage database (dff_t1.ucdb) to test1
    - Run simulation for the second testbench
    - Close the simulation (quit -sim) to save the coverage database (dff_t2.ucdb) to test2
    - Use this command to merge the 2 databases into one database (dff_merged.ucdb)
        - vcover merge dff_merged.ucdb dff_t1.ucdb dff_t2.ucdb -du dff
    - Generate a coverage report for the merged ucdb and analyze the coverage report

Deliverables:

One PDF file having the following:

1. Design file
    a. If the design has bugs, then fix them otherwise upload the given design file.
2. Testbench file
3. Do file

4. Coverage report text file
1. **Clear and neat** QuestaSim waveform snippets showing the functionality of the design with **each verification plan item**
2. Branch, statement and toggle coverage report snippets with justification if you could not reach 100% coverage for the designs.